The Erdős #249/#257 Lean formalization:

a machine-checked study of #249 and #257 — what is proved, what is cited, and what stays open

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Companion to the Lean 4 repository plectis-lean-erdos249-257, pinned at commit 4fb5915.

Abstract

This note describes, in ordinary mathematical language, a machine-checked Lean 4 development built around two Erdős irrationality problems: the irrationality of Erdős–Borwein-type series $\sum 1/(b^{a_k}-1)$ (Erdős #257), and the still-open question of whether the totient constant $S = \sum_{n\geq 1} \varphi(n)/2^n$ is irrational (Erdős #249). The development proves the Erdős–Borwein constant irrational for every integer base $b \geq 2$ and does the same for a calculus of named infinite supports; it recoordinatizes S as a weighted Erdős–Borwein object; it proves *unconditionally* that S equals no rational with denominator up to about 7.96×10^{34}; and it *reduces* the irrationality of S to a single sequence of finite, decidable certificates whose unboundedness is the untouched analytic core. It solves neither open problem, and it ships machine-checked non-claims that say so. Every formal statement is hyperlinked to the exact Lean source line, so a reader can move from a sentence of prose to the checked proof in one click. The Lean kernel verifies the whole package with no `sorry`, no `admit`, and no `native_decide`.

Contents

1 Main results and non-results

This section is the whole paper on one page: what the Lean development proves and, just as load-bearing, what it does not. Every item is machine-checked at the pinned commit; later sections expand each one and link it to source. The two named problems are Erdős #249 — is $S = \sum_{n\ge1} \varphi(n)/2^n$ irrational? — and Erdős #257 — is $\sum_{n\in A} 1/(2^n - 1)$ irrational for *every* infinite $A \subseteq \mathbb{N}$? Both are open, and this development solves neither.

The Mersenne–Lambert ladder. $L(f) = \sum_{n\ge1} \dfrac{f(n)}{2^n - 1}$

$L(\mu) = \frac{1}{2}$	$L(\varphi) = 2$	$L(\mathbf{1}) = E$	$L(\varphi * \mu) = S$ (#249)
rational	rational	**irrational (proved)**	**open**

The #249 reduction chain.

$$S \in \mathbb{Q} \implies \text{eventual binary periodicity} \implies R_{N+h} - R_N \in \mathbb{Z}$$

$$\implies \text{a finite } \mathsf{certifiedKill} \text{ witnesses failure} \iff R_{N+h} - R_N \notin \mathbb{Z}$$

(completeness: the finite certificate is exact, not a truncation heuristic;
#249 becomes “the certificate supply is unbounded”)

Proved (machine-checked).

- **All-base Erdős–Borwein irrationality.** $\sum_{n\ge1} 1/(b^n - 1)$ is irrational for every integer $b \ge 2$; the Erdős–Borwein constant E is the base-2 case (Result 4). This is a formalization of Erdős's classical theorem, and it is exactly the integer-base slice of Erdős #1049.
- **A #257 support calculus.** $\sum_{n\in A} 1/(b^n - 1)$ is irrational, at every base $b \ge 2$, for several structured infinite supports A: factorial, powers of two, multiples, pairwise-coprime with summable reciprocals, eventually periodic, residue classes, and the odd numbers (Result 5). These are named cases, not the universal #257 theorem.
- **A change of coordinates for #249.** The totient constant is a weighted Erdős–Borwein object, $S = L(\varphi * \mu)$ with a nonnegative primitive-conductor weight, and equals $\frac{1}{2} + \sum_d \mu(d)/(2^d - 1)^2$ and $\frac{1}{2} + \Pr(\gcd(X, Y) = 1)$ for independent fair-coin waiting times (Results 2, 3, 7).
- **An unconditional denominator exclusion.** S equals no rational of denominator up to about 7.96×10^{34} (Result 6). This is a record on the denominator, not a proof of irrationality.
- **A reduction of #249 to a finite, decidable supply.** Rationality of S forces an integer tail-period law; a finite certificate witnesses its failure, and by a completeness theorem the certificate exists *exactly* when the tail difference is non-integral (Results 8–11). The initial segment of the required supply is machine-checked non-empty (Result 12).

Not proved, and not claimed. The irrationality of S (#249) and the universal #257 statement are open. The reduction turns #249 into the single assertion that the certificate supply is *unbounded*; that unboundedness is the analytic core, and it is deliberately untouched. The repository states these boundaries in code as machine-checked non-claims (`not_erdos_249_solution`, `not_erdos_257_solution`; `NON_CLAIMS.md`).

How it is checked. The Lean 4 kernel verifies the whole package against Mathlib with no `sorry`, no `admit`, and no `native_decide`; every finite computation is discharged by `decide`. Section 9 gives the reproduction commands.

2 How to read this document

I wrote this so that a number theorist who has never used a proof assistant can see exactly what has and has not been proved, and can check any individual claim without reading Lean. Two conventions make that possible.

First, the mathematics is stated in ordinary notation. The Lean names are given only as pointers. Second, every formalized statement carries a link of the form `CertificateKernel.lean:8006` (`irrational_erdosBorwein_series`). The blue `file:line` text is a hyperlink to that line of source on GitHub, pinned to the commit `4fb5915`; the name in parentheses is the Lean declaration. Line numbers refer to that commit and do not drift.

I have tried to be exact about the boundary between what is proved, what is only cited, and what is open. The repository enforces that boundary in code through a set of non-claims (`NON_CLAIMS.md`), and I repeat it here in Section 8.

3 Two constants, two problems

Work at base 2 unless a base b is named. Two Dirichlet-type series carry the whole story.

- The **Erdős–Borwein constant** $E = \sum_{n\geq 1} \dfrac{1}{2^n - 1}$.
- The **totient constant** $S = \sum_{n\geq 1} \dfrac{\varphi(n)}{2^n}$, where φ is Euler's totient.

The #257 direction. Erdős #257 asks whether $\sum_{n\in A} 1/(2^n - 1)$ is irrational for *every* infinite $A \subseteq \mathbb{N}$. The development proves several genuine cases of the analogous statement $\sum_k 1/(b^{a_k} - 1)$, at every base $b \geq 2$, but not the universal statement.

The #249 direction. Whether S is irrational is open. The development contributes an unconditional exclusion of small-denominator rationals and a conditional reduction of the whole problem to finite certificates. Neither is a solution, and neither is presented as one.

The two constants are one convolution apart on a single ladder, which is the right place to start.

4 The Mersenne–Lambert ladder

Define the Lambert-type transform of an arithmetic function f by

$$L(f) \;=\; \sum_{n\geq 1} \frac{f(n)}{2^n - 1} \;=\; \sum_{m\geq 1} \frac{(f * \mathbf{1})(m)}{2^m}, \qquad (f * \mathbf{1})(m) = \sum_{d\mid m} f(d),$$

the second equality being the standard Lambert rearrangement (one Dirichlet convolution by the constant $\mathbf{1} = \zeta$). Walking the ladder by convolution lands on five values spanning rational, irrational, open, and transcendental character.

input f	$L(f)$	value	status in the repository
μ (Möbius)	$L(\mu)$	$1/2$	rational, proved
φ (totient)	$L(\varphi)$	2	rational, proved
$\mathbf{1}$ (constant one)	$L(\mathbf{1})$	E	**irrational, machine-checked**
$A = \varphi * \mu$	$L(A)$	S	**open (Erdős #249)**
Id	$L(\mathrm{Id})$	$\sum \sigma(m)/2^m$	transcendental (Nesterenko 1996 [4], cited only)

The weight $A = \varphi * \mu$ is the primitive-conductor count: $A(p) = p - 2$ on primes, $A \geq 0$, and $A * \zeta = \varphi$. So one convolution by ζ separates the open constant S from the rational 2, and one more separates rational from transcendental ($\varphi * \zeta = \mathrm{Id}$, $\mathrm{Id} * \zeta = \sigma$). This is the paper's organizing move: S is not an unrelated totient sum but the rung one convolution from the machine-checked irrational E, the same series $\sum(\cdot)/(2^d - 1)$ with the constant weight 1 replaced by the nonnegative weight A.

Result 1 (the rational rungs). *$\sum_{d\geq 1} \mu(d)/(2^d - 1) = 1/2$ and $\sum_{d\geq 1} \varphi(d)/(2^d - 1) = 2$, exactly.*

Formalized: `MersenneLambertLadder.lean:527` (`tsum_moebius_div_two_pow_sub_one_eq_half`) and `MersenneLambertLadder.lean:515` (`tsum_totient_div_two_pow_sub_one_eq_two`); restated on the #249 series shape at `tsum_moebius_div_two_pow_sub_one_eq_half` and `tsum_totient_div_two_pow_sub_one_eq_two`.

Result 2 (the positive lift, $L(A) = S$). *With the primitive-conductor weight $A = \varphi * \mu$,*

$$\sum_{d\geq 1} \frac{A(d)}{2^d - 1} \;=\; \sum_{n\geq 1} \frac{\varphi(n)}{2^n} \;=\; S.$$

Formalized: `CertificateKernel.lean:18116` (`tsum_primWeight_div_two_pow_sub_one_eq_totient_series`), with the weight A defined at `primWeight`. This places S in the same $\sum(\cdot)/(2^d - 1)$ family as E, with a nonnegative weight.

Result 3 (the Möbius-square lens).

$$S \;=\; \tfrac{1}{2} \;+\; \sum_{d\geq 1} \frac{\mu(d)}{(2^d - 1)^2}.$$

Formalized: `CertificateKernel.lean:18125` (`totient_series_eq_half_add_moebius_mersenne_square`). Writing $T = \sum_{d\geq 1} \mu(d)/(2^d - 1)^2$, irrationality of S is equivalent to irrationality of T, since $S = \frac{1}{2} + T$. The factor $1/(2^d - 1)^2$ is exactly the probability that d divides two independent fair-coin waiting times (Section 6.2), which is why this signed constant, the coprime-pair mass, and the tail differences of Section 6.3 are three faces of one object.

5 Erdős–Borwein-type irrationality (the #257 direction)

5.1 Full support: the Erdős–Borwein constant, every base

Result 4 (full-support irrationality). *For every integer $b \geq 2$, the series $\displaystyle\sum_{n\geq 1} \frac{1}{b^n - 1}$ is irrational. In particular the Erdős–Borwein constant $E = \sum_{n\geq 1} 1/(2^n - 1)$ is irrational.*

Formalized: `CertificateKernel.lean:7999` (`irrational_erdosSum_full_support`) (all bases) and `CertificateKernel.lean:8006` (`irrational_erdosBorwein_series`) (the base-2 corollary). This is Erdős's 1948 theorem [1]. In the ladder of Section 4 it is the rung $L(\mathbf{1})$, and it is the fixed machine-checked point one convolution away from the open S.

The proof runs through the certificate machine of the kernel rather than through a direct digit argument: a bounded Bertrand/CRT frame on a first block, a middle-window average over divisor pairs with a pigeonhole selection, and an explicit closure of the parameters. The Lambert bridge $\sum 1/(b^n - 1) = \sum_m \tau(m)/b^m$ (divisor-count series) is `erdosSum_full_support_eq_tsum_divisor_count` in the source.

5.2 Named infinite-support cases

Beyond full support, the development proves irrationality for a family of infinite supports A, at every base $b \geq 2$. Each is a genuine formalized case of the #257 statement family; none is the universal statement, and none is asserted to be new mathematics.

Result 5 (support-class family)**.** *For every integer $b \geq 2$, the series $\sum_{n\in A} 1/(b^n - 1)$ is irrational for each of the following supports A:*

(a) *factorial support $A = \{n! : n \geq 1\}$, giving $\sum 1/(b^{n!} - 1)$;*

(b) *power-of-two support $A = \{2^n : n \geq 0\}$, giving $\sum 1/(b^{2^n} - 1)$;*

(c) *multiples $A = d\mathbb{N}$;*

(d) *pairwise-coprime supports with summable reciprocals (Erdős's condition [2]);*

(e) *eventually-periodic supports, residue classes, and the odd numbers.*

These sort into three groups. The classical case is (d): infinite pairwise-coprime A with $\sum_{a\in A} 1/a < \infty$, formalized from Erdős's condition. The lcm-gap named cases are (a) and (b), whose support gaps outgrow the running lcm; the structured families are (c) and (e).[1] Formalized: (a) `irrational_erdosSum_factorial_support`; (b) `irrational_erdosSum_two_pow_support`; (c) `irrational_erdosSupportSeries_multiples`; (d) `irrational_erdosSupportSeries_pairwise_coprime`; (e) `irrational_erdosSupportSeries_eventuallyPeriodic`, `irrational_erdosSupportSeries_residueClass`, `irrational_erdosSupportSeries_odd`. The two base-2 instances that name the problem directly are `erdos257_family_factorial_instance` and `erdos257_family_two_pow_instance`.

The factorial and power-of-two cases are worth a word because they are not reachable by a Liouville shortcut: the power-of-two gap 2^k − lcm stays within a bounded ratio of the lcm, so the terms do not decay fast enough for a transcendence-measure argument, and a uniform criterion is needed. The mechanism behind these is what the repository calls *period noncollapse*: a prime-valuation-deficit witness forces the multiplicative order of the base modulo the relevant modulus not to collapse, which forbids the long repeated digit blocks a rational value would require.

Remark. A source comment near `lcm_gap_hypothesis_fails_full_support` describes a second, near-integer route to full support whose witnesses are not built there, and says so; the full-support theorem is proved by the separate certificate engine cited under Result 4. The two coexist; the near-integer route is a description of one lane, not a retraction of the theorem.

5.3 What stays open

The universal Erdős #257 statement, irrationality of $\sum_{n\in A} 1/(2^n - 1)$ for *every* infinite A, is not proved. The proved cases are a family, not the quantified theorem, and the repository marks this in code (the non-claim `not_erdos_257_solution` in `NON_CLAIMS.md`).

[1] Periodic-coefficient Lambert series have their own literature; see e.g. Luca–Tachiya [6]. The cases here are formalized in the repository's own framework and are not claimed as new results.

6 The totient constant S (Erdős #249)

Whether $S = \sum_{n\geq 1} \varphi(n)/2^n$ is irrational is open. The development gives two things and claims neither as a solution: an unconditional exclusion of small-denominator rationals, and a conditional reduction to finite certificates.

6.1 Unconditional: no small denominator

Result 6 (denominator exclusion). *For every integer p and every q with $1 \leq q \leq 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \approx 7.96 \times 10^{34}$,*

$$S \neq \frac{p}{q}.$$

Equivalently, if S is rational then its reduced denominator exceeds that bound.

Formalized: `CertificateKernel.lean:18055` (`tsum_totient_div_pow_two_ne_ratCast_of_den_le_79639646646701375323355774875831053`). The bound is reached by a ladder of finite, elementary exclusions,

$$4838 \;\to\; 4\,194\,304 = 2^{22} \;\to\; 2.49 \times 10^{17} \text{ (Farey } K{=}120) \;\to\; 7.96 \times 10^{34} \text{ (Farey } K{=}240),$$

each a mediant/Farey argument that traps any candidate m/q inside a forbidden gap. See `tsum_totient_div_pow_two_ne_ratCast_of_den_le_4838` for the first rung and `gap_check_window_1_120_le_248672326362367909`, `gap_check_window_1_240_le_79639646646701375323355774875831053` for the two Farey windows.

This is a record on the denominator, not a proof of irrationality. A rational with a larger denominator is not excluded by it.

Remark (one record, four faces). Because the ladder identities of Section 4 are equalities, the same finite record transfers verbatim to three other representations of the constant: to the positive lift $\sum_d A(d)/(2^d - 1)$ (`CertificateKernel.lean:18144`), to the signed Möbius-square constant $T = \sum_d \mu(d)/(2^d - 1)^2$ at half the bound (`CertificateKernel.lean:18158`), and to the coprimality probability of Section 6.2 at half the bound. Half appears because $S = \frac{1}{2} + T$ turns a denominator d for T into $2d$ for S.

6.2 The geometric picture: S as coprime-pair mass

Let X, Y be independent fair-coin waiting times, $\Pr(X = n) = 2^{-n}$ for $n \geq 1$. Then $\Pr(d \mid X) = 1/(2^d - 1)$, so the Lambert transform is a divisor calculus of X: $L(f) = \mathbb{E}[(f * \zeta)(X)]$. The rungs read as $L(\mu) = \Pr(X = 1) = 1/2$, $L(\varphi) = \mathbb{E}[X] = 2$, $L(\mathbf{1}) = \mathbb{E}[\tau(X)] = E$, and $L(A) = \mathbb{E}[\varphi(X)] = S$.

Counting visible lattice points (a, b) with $a + b = n$, $a > 0$, $\gcd(a, b) = 1$ gives exactly $\varphi(n)$, uniformly in n, so S is itself a coprime-pair mass.

Result 7 (coprimality probability).

$$S \;=\; \tfrac{1}{2} \;+\; \Pr\bigl(\gcd(X, Y) = 1\bigr) \;=\; \tfrac{1}{2} \;+\; \sum_{\substack{a,b\geq 1\\ \gcd(a,b)=1}} 2^{-(a+b)}.$$

Formalized: `CertificateKernel.lean:18228` (`totient_series_eq_half_add_visible_coprime_pairs`), with the underlying lattice identity at `tsum_visible_coprime_pairs_eq_totient_series`. Base 2 is the one point where $\Pr(X = a)\Pr(Y = b) = 2^{-(a+b)}$ needs no normalizing constant, which is why the totient sum and the coprimality probability line up with no boundary correction. Erdős #249 is, in this reading, the question of whether two independent fair-coin waiting times are coprime with irrational probability.

6.3 Conditional: reduction to finite certificates

The reduction turns irrationality of S into the non-vanishing of a decidable predicate at infinitely many parameters. Everything in it is elementary: $\varphi(n) \le n$, geometric tails, and integer window arithmetic. There is no q-Padé input, no Stern–Brocot input, and no unformalized citation (`TotientTailPeriodKiller`).

Step 1: rationality forces an integer tail-period law. Write the fractional layer of $2^N S$ as the tail

$$R_N \;=\; \sum_{m\ge 1} \frac{\varphi(N+m)}{2^m}.$$

If S is rational, its binary expansion is eventually periodic, which forces, for a suitable period h and all large N, the integrality $R_{N+h} - R_N \in \mathbb{Z}$. To refute integrality at a finite depth L, form the window discrepancy

$$\Delta_{h,N,L} = \sum_{j=0}^{L-1} \big(\varphi(N+h+1+j) - \varphi(N+1+j)\big)\, 2^{\,L-1-j} \in \mathbb{Z},$$

the first L binary digits of $R_{N+h} - R_N$ up to a controlled deep tail (bounded using only $\varphi(n) \le n$), and call the pair *certified* when the residue of $\Delta_{h,N,L}$ modulo 2^L misses a neighbourhood of 0:

$$\mathsf{certifiedKill}(h,N,L):\quad N+h+L+2 \;<\; \big(\Delta_{h,N,L} \bmod 2^L\big) \;<\; 2^L - (N+h+L+2).$$

This is a finite integer computation, decidable by the kernel. It is the natural finite object here precisely because a rational S would pin the tail difference to an integer, and a residue that stays a fixed distance from 0 is a finite proof that it did not.

Result 8 (tail-period reduction)**.** *If for every period $h \ge 1$ and every threshold N_0 there exist $N \ge N_0$ and L with $\mathsf{certifiedKill}(h,N,L)$, then S is irrational.*

Formalized: `TotientTailPeriodKiller.lean:386` (`irrational_totient_series_of_certificate_supply`); the tail R_N and the predicate are `totientTail` and `certifiedKill`.

Step 2: collapse the two parameters to one. Every multiple of a period is a period, so $\mathrm{lcm}(1,\dots,t)$ is the universal-period envelope: any eventual period of a rational S divides it once t is large, so standing on that one diagonal ray dominates all possible periods at once, and the position parameter drops out too.

Result 9 (diagonal reduction)**.** *Write $\mathrm{lcm}(1..t)$ for $\mathrm{lcm}(1,\dots,t)$. If for every t_0 there exist $t \ge t_0$ and L with $\mathsf{certifiedKill}\big(\mathrm{lcm}(1..t), \mathrm{lcm}(1..t), L\big)$, then S is irrational.*

Formalized: `LcmDiagonalReduction.lean:92` (`irrational_totient_series_of_lcm_diagonal_certificate_supply`).

Step 3: the cone, flatness, and completeness. On the cone $\{k \cdot \mathrm{lcm}(1..t)\}$, rationality is spent to its limit: it forces not one pairwise relation but a single fractional constant across the whole cone.

Result 10 (cone flatness)**.** *If S is rational, then the relevant tail values share one fractional part across the whole* lcm *cone.*

Formalized: `LcmConeFlatness.lean:90` (`rational_totient_series_forces_lcm_cone_flatness`).

The certificates are not a lossy heuristic: a certificate exists exactly when the object it certifies is non-integral. This is the theorem that turns the finite computations from experiments into receipts.

Result 11 (certificate completeness)**.** *For all h, N,*

$$\big(\exists L,\ \mathsf{certifiedKill}(h,N,L)\big) \iff R_{N+h} - R_N \notin \mathbb{Z}.$$

Formalized: `LcmConeFlatness.lean:316` (`exists_certifiedKill_iff_tail_diff_notMem_int`). Finite certificates are receipts; non-integrality is the object. A whole menu of cone vertices can be interrogated at once by a joint refuter that is sharper than any pairwise test: it refutes the single-fractional-part model that rationality predicts on a finite sample of cone vertices (`LcmConeNonflat.lean:126` (`exists_nonintegral_pair_of_coneNonflatCert`)).

The certificates are non-empty. The initial segment of the required supply is machine-checked, so the reduction is not vacuous. By completeness, each of these is not a numerical experiment but a kernel-checked proof of a specific non-integral tail difference.

Result 12 (finite deposits)**.** *The following are checked by the Lean kernel:*

- $\mathsf{certifiedKill}(h, 12, 16)$ *for every period* $h \in [1, 8]$*;*
- *certified kills for every period* $h \in [1, 16]$ *at a fixed window;*
- *diagonal certificates* $\mathsf{certifiedKill}(\mathrm{lcm}(1..t), \mathrm{lcm}(1..t), L)$ *for* $t \in [1, 6]$*, and at* $t = 7, 8$*;*
- *a tabulated joint cone-vertex certificate.*

Formalized: `certifiedKill_all_small`, `certifiedKill_all_upto_sixteen`, `certifiedKill_periodLcm_diagonal_upto_six` (with `certifiedKill_periodLcm_diagonal_at_seven`, `certifiedKill_periodLcm_diagonal_at_eight`), and `coneNonflatCert_cells`. These deposits prove that each layer of the reduction is non-vacuous and, by completeness, exact; they are not evidence that the unbounded supply exists.

What is open. The reduction converts #249 into the statement that the certificate supply is *unbounded*: certificates exist for arbitrarily large parameters. By completeness (Result 11) this is equivalent to the tail differences being non-integral infinitely often, and equivalently to the Farey gap denominators of Section 6.1 being unbounded — the same obstruction seen as a rational denominator, an eventual binary period, and a flat tail on the lcm cone. That unboundedness is the analytic core, and it is deliberately untouched. The reduction is real; the closure is not asserted.

7 Where this sits: the Erdős landscape

The table records how the development relates to the numbered problems it is near. Statuses are as listed on the individual problem pages erdosproblems.com/{249, 257, 1049, 69, 250, 258} in July 2026; none of the external results is re-proved here.[2]

[2] The irrationality of S (#249) and the universal #257 statement are open. #1049 is open for rational $t > 1$, with the integer-base case attributed to Erdős. #69 (Tao–Teräväinen [7]), #250 (Nesterenko 1996 [4]), and #258 (recorded as proved in Lean) are solved; none is re-proved here.

problem	status	relation in this repository
#249 $\sum \varphi(n)/2^n$	open	the target; unconditional denominator exclusion and a conditional reduction, no solution
#257 $\sum_{n\in A} 1/(2^n - 1)$	open	named infinite-support cases and the full-support ($A = \mathbb{N}$) case, not the universal statement
#1049 $\sum 1/(t^n - 1)$, $t \in \mathbb{Q}_{>1}$	open (rational t)	the integer-base case $b \geq 2$ is formalized (Result 4); the rational case is untouched
#69 $\sum \omega(n)/2^n$	solved [7]	the prime-support case of #257; only the identity bridge to $\sum_p 1/(2^p - 1)$ is formalized, not the irrationality
#250 $\sum \sigma(n)/2^n$	solved [4]	the ladder neighbour $L(\mathrm{Id})$; cited, not reproved
#258 $\sum \tau(n)/(a_1 \cdots a_n)$	solved	not addressed here; the monotone φ/σ sequel is the sibling territory of #249

The honest one-line summary: this is a formalization of Erdős–Borwein-type irrationality together with an unconditional record and a finite-certificate reduction for the open totient constant. It is not a solution to #249 or to the universal #257.

8 What is proved, cited, and open

Proved (machine-checked, no `sorry`, no `admit`). Irrationality of $\sum 1/(b^n - 1)$ for every $b \geq 2$ and of the named support cases (Results 4, 5); the ladder identities and the Möbius-square and coprimality reformulations (Results 2, 3, 7); the unconditional denominator exclusion to 7.96×10^{34} (Result 6); the full chain of conditional #249 reductions with certificate completeness and a non-empty initial supply (Results 8–12).

Cited, not formalized here. Transcendence of $\sum \sigma(m)/2^m$ (Nesterenko 1996 [4]) and the q-zeta anchor (Postelmans–Van Assche [5]). They are named as ladder neighbours.

Open, and not solved here. Irrationality of S (#249); the universal #257 statement. The conditional reductions turn #249 into "the certificate supply is unbounded", which is the untouched analytic core.

The repository states these boundaries in code as machine-checked non-claims: `not_erdos_249_solution`, `not_erdos_257_solution`, and four provenance non-claims (`NON_CLAIMS.md`, `PlectisSnapshot/PublicAPI.lean`).

9 Verification and reproducibility

The whole package is checked by the Lean 4 kernel against Mathlib, pinned in `lake-manifest.json` at toolchain `leanprover/lean4:v4.29.1`. Three facts matter for trust.

- There is no `sorry` and no `admit` anywhere in the package.
- Every finite computation is discharged by `decide`, checked by the kernel. There is no use of `native_decide`, so no compiler is trusted.
- The package has more than 1,600 theorem declarations. The assembled microkernel is `CertificateKernel.lean` (about 18,900 lines); the machine-generated finite certificates are `GeneratedCertificates.lean` (about 27,700 lines) plus per-base shards, each instance closed by `decide`.

To build and check:

```
lake exe cache get      # fetch the prebuilt Mathlib cache
lake build              # build and kernel-check the package
python3 scripts/verify_snapshot.py --allow-blocked   # structural verify
```

A standalone verifier `scripts/verify_snapshot.py` (Python standard library only) recomputes the tree digest against `RELEASE_PROVENANCE.json` and checks the license, the SPDX/REUSE metadata, the public-API adapter, and the required non-claims; it reports `structural_status: pass`. It is a local-candidate structural check, not a publication authority: its `status: blocked` marker is a two-repo release-topology gate that stays set until a separate promotion lane attaches the exact public problem-numbering citation, and it is independent of the Lean build the kernel checks in full. Provenance and context for how this snapshot was produced — and its relation to the private system that generated it — are in the repository README.

10 A sibling project, in progress

Separately from this repository, I am working on the Ramanujan Machine Challenge (ten problems, released July 2026, on proofs or symbolic derivations of explicit formulas for constants such as π, e, Catalan's constant, and zeta values). That work is a different Lean lane and is not part of this snapshot: no challenge target is claimed solved or formalized here, and nothing in this paper depends on it. A coarse status map for that project lives in the repository README.

A Index of principal declarations

Each row links to the declaration at commit `4fb5915`.

statement	source line (and declaration)
$\sum 1/(b^n-1)$ irrational, all $b \ge 2$	CertificateKernel.lean:7999 `irrational_erdosSum_full_support`
Erdős–Borwein E irrational	CertificateKernel.lean:8006 `irrational_erdosBorwein_series`
factorial support	CertificateKernel.lean:5706 `irrational_erdosSum_factorial_support`
power-of-two support	CertificateKernel.lean:5730 `irrational_erdosSum_two_pow_support`
pairwise-coprime supports	CertificateKernel.lean:10447 `irrational_erdosSupportSeries_pairwise_coprime`
$L(\mu) = 1/2$	MersenneLambertLadder.lean:527 `tsum_moebius_div_two_pow_sub_one_eq_half`
$L(\varphi) = 2$	MersenneLambertLadder.lean:515 `tsum_totient_div_two_pow_sub_one_eq_two`
positive lift $L(A) = S$	CertificateKernel.lean:18116 `tsum_primWeight_div_two_pow_sub_one_eq_totient_series`
Möbius-square lens	CertificateKernel.lean:18125 `totient_series_eq_half_add_moebius_mersenne_square`
$S = \frac{1}{2} + \Pr(\gcd = 1)$	CertificateKernel.lean:18228 `totient_series_eq_half_add_visible_coprime_pairs`
$S \neq p/q$ for $q \le 7.96 \times 10^{34}$	CertificateKernel.lean:18055 `tsum_totient_div_pow_two_ne_ratCast_of_den_le_7963964664670137532335577487583 1053`
tail-period reduction	TotientTailPeriodKiller.lean:386 `irrational_totient_series_of_certificate_supply`
diagonal reduction	LcmDiagonalReduction.lean:92 `irrational_totient_series_of_lcm_diagonal_certificate_supply`
cone flatness from rationality	LcmConeFlatness.lean:90 `rational_totient_series_forces_lcm_cone_flatness`
certificate completeness	LcmConeFlatness.lean:316 `exists_certifiedKill_iff_tail_diff_notMem_int`
joint cone refuter	LcmConeNonflat.lean:126 `exists_nonintegral_pair_of_coneNonflatCert`
non-empty certified-kill segment	TotientTailPeriodKiller.lean:396 `certifiedKill_all_small`

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